

Problem 302: Presidential Escort

Difficulty: Hard

Author: Adam Rigdon, Owego, New York, United States

Originally Published: Code Quest 2026

Problem Background

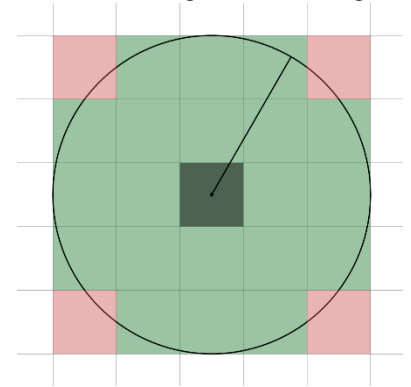
This past August, The U.S. Marine Corps formally accepted the 23rd and final next-generation VH-92A presidential helicopter built by Sikorsky, a Lockheed Martin Company. As you might expect, this presidential helicopter is built to be extremely safe and reliable, given the nature of its precious cargo.

You're working with the Marine Corps' safety team to ensure that the President's time spent in the helicopter is as safe as possible, regardless of its involvement in dangerous situations.

Problem Description

The President has been very involved in recent military conflict and has been much nearer to the battlefield than usual. Unfortunately, the Marine Corps have received information that their location has been compromised, and enemy aircraft are on their way. They need your help to find a safe path further from the front lines while avoiding detection from the many radar bases on the ground along the way.

Each radar base has a given radius that extends from the center of its location to create a set of coordinates in the grid that it can detect. You can imagine that this grid is created by drawing a circle with the given radius (plus 0.5) at the center of the base's coordinate. Any other grid square whose center falls within the circle is one that the sonar base can detect. The image at right demonstrates this for a radar base with a detection radius of 2 spaces; the red spaces are outside the detection range for this base.



You will be given a 2-dimensional grid that represents a top-down view of the President's area. Each space within the grid contains a character showing what's at that location:

- A period (.) indicates an open space.
- An uppercase letter P indicates the President's current location, which will not be within the detection radius of any radar base.
- An uppercase letter D indicates the President's destination, which will not be within the detection radius of any radar base.
- Numbers (1-9) indicate the presence of an enemy radar base; the exact number indicates the base's detection radius.

You must find the length of the shortest path from the President's current location to the destination that avoids detection. Note that the President's convoy can only move in cardinal directions (that is, not diagonally). It is guaranteed that there is a solution for each scenario.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include:

- A line containing two positive integers, separated by spaces:
 - The first integer, **X**, represents the number of columns in the grid
 - The second integer, **Y**, represents the number of rows in the grid
- **Y** lines containing the layout of the grid. Each line consists of **X** characters, which may include periods, uppercase letters P or D, and/or numeric digits from 1-9 inclusive, as described above. Each grid will contain exactly one each of the uppercase letters P and D.

```
1
7 5
.P.....
....2...
.....
.....D
.....
```

Sample Output

For each test case, your program must print a single line containing an integer, representing the length of the shortest path from P to D. Include the destination, but not the starting point, in this count.

```
10
```